Spectral–Spatial Prototype Learning-Based Nearest Neighbor Classifier for Hyperspectral Images

Dan Li, Yuan Shen, Fangqiang Kong, Jiahang Liu, *Member, IEEE*, and Qiang Wang, *Member, IEEE*

***Abstract*— Due to the Hughes phenomenon, hyperspectral image (HSI) classification under small sample size situation is still a key challenging problem. To alleviate this issue, we propose a novel spectral–spatial prototype learning-based nearest neighbor classifier (SSPLNN) for HSI in this article. The local spectral–spatial neighbor set is first constructed for each sample based on both spectral similarity and spatial structural context to accurately explore the local spectral–spatial information. Then, a spectral–spatial prototype learning model is designed to learn a set of spectral–spatial prototypes, which can optimally utilize both the similarity and variance of samples within each spectral–spatial set and excavate the unseen spectral–spatial variations. The learned spectral–spatial prototypes offer more complementary information to improve the classification accuracy remarkably under small sample size situation. In addition, a linear discriminative projection is simultaneously learned to make each test local spectral–spatial set to be optimally classified to the same class with its nearest neighbor (NN) spectral–spatial prototype set in the projected target subspace. Finally, the NN classifier based on measuring the minimum geometric distance between the projected test spectral–spatial set and the projected spectral–spatial prototype sets is employed to determine the label. Experimental results demonstrate that the proposed SSPLNN method outperforms several well-known classification methods by a large margin on three widely analyzed HSI datasets.**

***Index Terms*— Feature extraction (FE), hyperspectral classification, nearest neighbor (NN), small size samples situation, spectral–spatial prototype learning.**

I. Introduction

HYPERSPECTRAL image (HSI), containing hundreds of continuous and narrow spectral bands, precisely characterizes the physical properties and plays a very important role in different applications, such as land-cover analysis [1], [2], precision agriculture [3], military target detection [4], and environmental monitoring [5]. Among these research fields, land cover classification aiming to assign a unique label for each test pixel on the whole HSI is one of the most important topics. However, HSI classification always imposes three main challenges. The first is that the high-dimensional data would arise difficulties to analyze and organize in high-dimensional spaces. The second is that the Hughes phenomenon [6] degrades the classification accuracy significantly under the small sample size situations. The third is that the existence of mixed spectral pixels would introduce noise to the classification maps.

In recent years, a large number of HSI classification methods [7], [8], [9], [10] have been proposed to address these three main challenges. Spectral–spatial FE methods [11], [12], [13] attempt to delve various discriminative spatial and spectral features to characterize the original samples and enhance the classification results. In [14], [15], and [16], a series of FE methods inspired by the extended morphological profile (EMP) are developed to extract various spatial contextual information adaptively, which can significantly improve the classification accuracy. Also, the local binary patterns (LBPs) and the Gabor are employed to extract the different spatial features, respectively, in [17], [18], [19], [20] and [21], [22], [23]. Hong et al. [24] utilize invariant attribute profiles to locally extract invariant features from HSI in both spatial and frequency domains. Moreover, some multiple spectral–spatial feature fusion techniques are presented in [25], [26], [27], and [28], which can sufficiently describe the pixels from different perspectives and further improve classification accuracy. However, since the extracted spectral–spatial features significantly affect the classification results, there is no guarantee that they can accomplish the best performance due to the different structures of different HSI datasets. Moreover, many ways to try to solve the intraclass difference and interclass similarity in HSI are developed to further improve the classification performance. A discriminative marginalized least-squares regression (DMLSR) method is proposed in [29] to learn a more discriminative projection matrix with consideration of class separability and data-reconstruction ability simultaneously, which can avoid the overfitting problem and enhance class separability. Yang et al. [30] develop a novel deep similarity network for HSI classification to overcome subtle interclass difference and large intraclass variability. Liu et al. [31] present an unsupervised dimensionality reduction method called superpixelwise collaborative representation graph embedding, which can simultaneously alleviate the Hughes phenomenon and the problem of intraclass difference and interclass similarity in HSI. Nevertheless, these methods are hard to achieve high classification accuracy when the available training samples are scarce. Extracting a suitable

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set of discriminative spectral–spatial features for HSI datasets with different structures and achieving good classification performance especially under the small sample size situations are arduous tasks in high-dimensional space.

Deep learning [32], [33], which aims to automatically extract "the most suitable features" based on different structures of the original datasets via a series of hierarchical layers, has been widely utilized for HSI classification. Convolutional neutral network (CNN)-based HSI classification methods [34], [35], [36], [37] are the most popular techniques and have demonstrated remarkable classification performance. Liu et al. [38] propose a content-guided CNN to suppress the irregularity and unexpected features in class boundaries and to improve the feature learning in cross-class regions, which possesses great details' preserving ability. An attention-aided CNN model incorporating attention modules to aid networks that focus on more discriminative channels or positions is presented in [39] for HSI classification. Feng et al. [40] present a novel attention multibranch CNN method based on adaptive region search, which can not only extract joint spatial-spectral features but also exploit complementary information among different layers. A deep prototypical network with hybrid residual attention is proposed in [41] to effectively investigate the spectral–spatial information and to obtain more superior results. Xi et al. [42] propose a unified multidirection network, where 3-D squeeze-and-excitation residual blocks are beneficial to enhance the model's generalization capability even with limited training samples. The active learning and CNN are integrated into a unified framework to construct an active deep learning approach for HSI classification in [43]. A novel backbone network called SpectralFormer is developed in [44] to learn spectrally local sequence information from neighboring bands of HSI and yield groupwise spectral embedding. Moreover, graph convolutional networks (GCNs) [45], [46], [47] have been recently presented and successfully applied in HSI classification. In [48], a fully connected graph is adaptively constructed, where label propagation serves as regularization to assist the graph network in learning proper edge weights that lead to improved classification performance. Ding et al. [49] present a graph convolution with adaptive filters and aggregator fusion to deal with the problems of land cover discrimination and spatial feature learning and to achieve superior experimental results. Adversarial prototype learning is proposed in [50] and applied in a deep learning framework to alleviate overfitting, to align the feature distributions between the training and testing sets, and to achieve state-of-the-art performance. Nevertheless, a large number of samples are needed to accurately train the deep networks, while a collection of a large number of labeled data is an expensive and time-consuming process, thereby rendering unsatisfactory classification performance under small sample size situations. To alleviate this problem, a large number of semi-supervised classifications [51], [52] have been developed for HSI. A semi-supervised method called iterative multitask regression is developed in [53] to effectively exploit the labeled and unlabeled data to improve the classification accuracy. Hong et al. [54] develop joint and progressive subspace analysis to learn a high-level, semantically meaningful, joint spatial-spectral feature representation from HSI. Kotzagiannidis and Schonlieb [55] present a novel semi-supervised superpixel-based framework with multifeature graph learning for HSI classification in light of a very limited amount of labeled data. A cross-scale graph prototypical network is developed in [56] to achieve semisupervised high-quality classification accuracy. In addition, few-shot HSI classification methods handling the small sample size problem have been proposed. A novel few-shot learning framework for HSI classification with class-covariance metric is presented in [57] to further enhance the performance with few labeled samples. However, deep learning architecture always contains a huge number of tunable parameters needed to be tuned, which leads to high computational complexity. Consequently, learning a suitable set of discriminative spectral–spatial features based on different geometric structures of HSI datasets to improve classification performance under small sample size situations is thus considered a challenging task.

Since affine hull (AH) model [58] covers the affine combinations of samples in the training sets, it is naturally considered to be used to generate synthesized samples to expand the training set under small sample size situation. Also, the AH model can extract both the similarity and variance of pixels within each set, and it is appropriate to be employed to adaptively represent the characteristics for each set of samples. In [59], a set-to-set distance-based spectral–spatial HSI classification method (SD-SSC) is proposed, where the similarity between training and test sets can be transformed to the geometric distance between the corresponding AHs. As the AH-based representation of sets can account for the similarity, variation, and some "virtual" semantic relationship between sample sets, SD-SSC achieves superior classification performance. Nevertheless, the original AH model generates over-large affine hull and is too loose, which leads to noise classification maps when there are mixed spectral pixels in the sample sets. In addition, the theoretical inefficiency of the set-to-set matching-based [60], [61] nearest neighbor (NN) classification method is unbearable in large-scale deployments and practical applications.

In this article, we propose a novel spectral–spatial prototype learning-based NN classifier (SSPLNN) for HSI. First, a local spectral–spatial neighbor set is constructed for each sample in the original HSI by exploiting both the spectral similarity and the spatial structural context information, which can accurately explore the local spectral–spatial information. Then, we design a spectral–spatial prototype learning model to learn a set of discriminative spectral–spatial features for each training spectral–spatial neighbor set, namely the spectral–spatial prototypes. Since the learned spectral–spatial prototypes are also synthesized by the linear combinations of existing samples in the set, they inherit the favorable property of AH model that utilizes both the similarity and variance of pixels within each spectral–spatial set to adaptively characterize the set. Moreover, they shrink and optimize the corresponding AH approximation discriminatively to make themselves more specifically suitable for characterizing each spectral–spatial set, which can excavate more unseen complementary spectral–spatial variations and improve the classification accuracy remarkably

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under small sample size situations. In addition, a linear discriminative projection is simultaneously learned to make each test local spectral–spatial set to be optimally classified to the same class with its NN spectral–spatial prototype set, which can further enhance the classification performance. Finally, the simplest NN classifier by measuring the minimum geometric distance between the projected test spectral–spatial set and the optimal projected spectral–spatial prototype sets is employed to determine the label of the test pixel. Experiments are implemented on three HSI datasets and the results demonstrate that our proposed SSPLNN method is superior to several state-of-the-art classification methods by a large margin especially under small sample size situations.

The major contributions of this article are summarized as follows.

1) We construct local spectral–spatial set for each sample to explore the local spectral–spatial information accurately.
2) We design spectral–spatial prototype learning model to simultaneously learn spectral–spatial prototypes optimally excavating the unseen spectral–spatial variations and linear discriminative projection to improve the classification accuracy remarkably under small sample size situation.
3) We employ the simplest NN classifier by measuring the minimum geometric distance between the projected test spectral–spatial set and the optimal projected spectral–spatial prototype sets to complete the classification task.

The remainder of this article is organized as follows. A review of the related works, e.g., the affine hull model, is presented in Section II. Section III presents the details of the proposed SSPLNN method. The experimental results from three benchmark HSI datasets are briefly described and analyzed in Section IV. Section V concludes this work.

II. Related Works

A. Affine Hull Model

Affine hull model is a simple but efficient way to model the image set, which covers the affine combinations of samples in the original image set to characterize the implicit semantic relationship between the existing samples in the image set. Moreover, the affine hull model can effectively complement the unseen appearance variations that even do not appear in the original image set. Let $\mathbf{X} = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_n\}$ represent an image set with n samples, where $\boldsymbol{x}_i, i = 1, 2, \ldots, n$ denotes the d-dimensional vector of the ith image. Then the affine hull model of the image set $\mathbf{X}$ can be formulated as follows:

$$\mathrm{AH}(\mathbf{X}) = \left\{ \boldsymbol{x} = \sum_{i=1}^{n} \boldsymbol{\alpha}_i \boldsymbol{x}_i, \quad \text{s.t.} \sum_{i=1}^{n} \boldsymbol{\alpha}_i = 1 \right\} \tag{1}$$

where $\boldsymbol{\alpha}_i$ is the corresponding coefficient to the sample $\boldsymbol{x}_i$. Thus, the AH model can be set as a valid descriptor of the image set and gives a rather loose approximation to the samples.

By using the mean vector $\boldsymbol{\mu}$ of all the samples, calculated by $\boldsymbol{\mu} = ((\sum_{i=1}^{n} \boldsymbol{x}_i)/n)$, as a reference to represent the image set, the affine hull model can be rewritten as follows:

$$\mathrm{AH}(\mathbf{X}) = \{\boldsymbol{x} = \boldsymbol{\mu} + \mathbf{U}\boldsymbol{v} / \boldsymbol{v} \in \mathbb{R}^l\} \tag{2}$$

where $\mathbf{U}$ is an orthonormal basis obtained by applying the singular value decomposition (SVD) to the data matrix $\{\boldsymbol{x}_1 - \boldsymbol{\mu}, \boldsymbol{x}_2 - \boldsymbol{\mu}, \ldots, \boldsymbol{x}_n - \boldsymbol{\mu}\}$, and $\boldsymbol{v}$ is a vector of parameters for the points within the subspace. Since we discard the directions corresponding to near-zero singular values, the dimension l of singular vectors in $\mathbf{U}$ is much smaller than the number of samples n. The AH model contains all the affine combinations of existing samples in the image set and can account for the unseen data similarity, variations with samples in image set.

III. Proposed Method

Fig. 1 shows the schematic of the proposed SSPLNN method for HSI. The main steps of SSPLNN are described briefly in the following sections.

A. Local Spectral–Spatial Neighbor Set

To accurately explore the spectral–spatial information, the local spectral–spatial neighbor set is constructed for each pixel in the original HSI dataset. Suppose $\mathcal{X} \in \mathbb{R}^{s_1 \times s_2 \times d}$ denotes the original HSI dataset, the details of how to generate the local spectral–spatial neighbor set are given as follows.

1) *Dimensional Reduction:* The maximum noise fraction (MNF) method [62] is first to implement on the original HSI dataset to obtain the dimension-reduced data $\mathcal{Z} \in \mathbb{R}^{s_1 \times s_2 \times r}$, which aims to reduce the computational complexity and remove the noise in the original HSI. The parameter r represents the reduced spectral dimension.
2) *Local Spatial Neighbor Set:* As superpixels [63], [64] can more adaptively explore the spatial information based on different types of spatial structures than square windows, we utilize superpixels to generate local spatial neighbor set. First, the first three principal components are extracted from the original HSI as base images by implementing principal component analysis (PCA). Then, the number of superpixels L is estimated by applying a texture detector (e.g., Sobel filter [65]) on base images. Next, we employ an oversegmentation algorithm [66] to base images to obtain the 2-D superpixel map. Finally, the indexes of the 2-D superpixel map are mapped to the dimension-reduced data $\mathcal{Z} \in \mathbb{R}^{s_1 \times s_2 \times r}$ to obtain the 3-D superpixels. For each sample $z_i, i = 1, 2, \ldots, n$, the local spatial neighbor set is constructed by all the samples in its corresponding superpixels and represented as $\mathbf{Z}_i = [z_i, z_i^1, \ldots, z_i^{m-1}]$, where m denotes the number of samples in superpixels.
3) *Local Spectral–Spatial Neighbor Set:* To further enhance the similarity of samples within the local spatial neighbor set, we identify the local spectral–spatial neighbors based on spectral similarity by choosing the M most similar neighbors based on cosine distance, which can be calculated by the following:

$$\cos\left(z_i, z_i^j\right) = \frac{\left\langle z_i, z_i^j \right\rangle}{\|z_i\|_2 \left\| z_i^j \right\|_2}, \quad j = 1, 2, \ldots, m-1. \tag{3}$$

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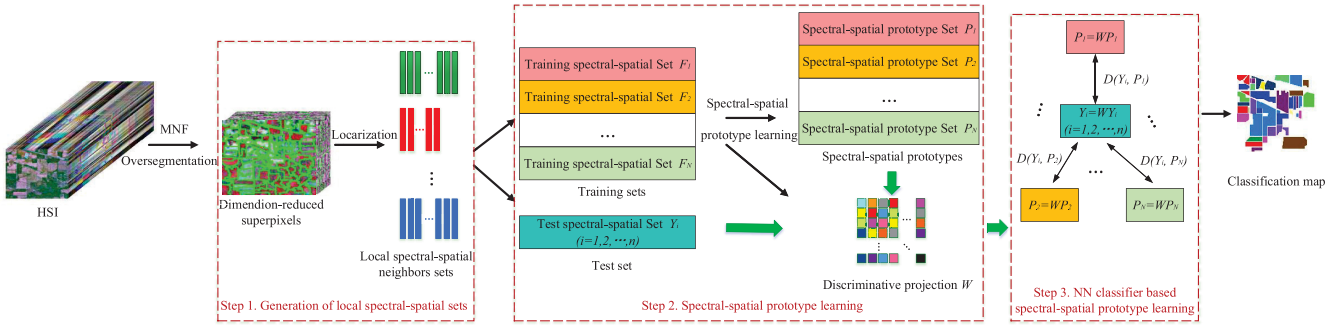


Fig. 1. Schematic of our proposed method SSPLNN for HSI classification.

Consequently, for each sample $z_i, i = 1, 2, \ldots, n$, the local spectral–spatial neighbor set is represented as $\mathbf{F}_i = [\boldsymbol{f}_i^1, \ldots, \boldsymbol{f}_i^M]$ based on both spectral similarity and spatial structural information, which can explore the spectral–spatial information accurately.

B. Spectral–Spatial Prototype Learning

In this section, we design a spectral–spatial prototype learning model to learn a set of discriminative spectral–spatial prototypes to characterize each training local spectral–spatial set. The learned spectral–spatial prototypes can optimally utilize both the similarity and variance of pixels within each spectral–spatial set and excavate the unseen spectral–spatial variations to offer more complementary information for better classification performance. Simultaneously, a discriminative linear projection is also learned to make each test local spectral–spatial set to be optimally classified to the same class with its NN spectral–spatial prototypes set in the projected target subspace. The learned linear projection can solve the problem that the spectral–spatial prototypes may not be sufficient to be discriminated linearly and further enhance the classification performance.

1) Training Spectral–Spatial Sets: As mentioned earlier, the local spectral–spatial neighbor set is constructed for each sample in the dimension-reduced HSI dataset to accurately explore the spectral–spatial information. In supervised classification methods, we first need to choose some samples with their correct labels to constitute a training set. Here, we randomly selected some pixels $\boldsymbol{f}_i, i = 1, 2, \ldots, N$ from all pixels that cover all classes as the training samples, where N represents the total number of training samples. Then, for each pixel $\boldsymbol{f}_i$, its local spectral–spatial neighbor set can be represented as $\mathbf{F}_i = [\boldsymbol{f}_i^1, \ldots, \boldsymbol{f}_i^M]$. Finally, the training spectral–spatial sets are constructed by combining the local spectral–spatial neighbor sets of the training samples, which can be represented as $\mathcal{F} = [\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_N]$.

2) Modeling of Spectral–Spatial Prototype Learning: To inherit the favorable property of AH model that utilizes both the similarity and variance of pixels within each spectral–spatial set, the spectral–spatial prototypes are also synthesized by the linear combinations of existing samples in the set. We simultaneously optimize the AH approximation discriminatively to make themselves more specifically suitable for characterizing each spectral–spatial set and excavate more unseen complementary spectral–spatial variations to improve the classification accuracy. For each local spectral–spatial set $\mathbf{F}_i = [\boldsymbol{f}_i^1, \ldots, \boldsymbol{f}_i^M], i = 1, 2, \ldots, N$, a spectral–spatial prototypes set $\mathbf{P}_i = [\boldsymbol{p}_i^1, \ldots, \boldsymbol{p}_i^l], i = 1, 2, \ldots, N$ can be calculated as follows:

$$\boldsymbol{p}_i^j = \boldsymbol{\mu}_i + \mathbf{U}_i \boldsymbol{v}_i^j, \quad j = 1, 2, \ldots, l. \tag{4}$$

At the same time, a discriminative linear projection $\mathbf{W}$ making each local spectral–spatial set to be optimally classified to the same class with its NN prototypes set in the projected target subspace is also learned. For each sample $\boldsymbol{f} \in \mathbf{F}_i$, its projection in the target subspace can be denoted as follows:

$$\boldsymbol{g} = \mathbf{W}^\top \boldsymbol{f}, \quad \boldsymbol{g} \in \mathbb{R}^k. \tag{5}$$

To achieve the goal that simultaneously learning the spectral–spatial prototypes to explore more discriminative spectral–spatial features and the linear projection to drive each local spectral–spatial set is very closer to its NN spectral–spatial prototype set from the same class, we define a loss function to model the spectral–spatial prototype learning as the following steps.

Step 1: For each sample $\boldsymbol{g}$ in the projected target subspace, its NNs, respectively, selected from the corresponding spectral–spatial prototypes of the same class and that of the different classes are first found, respectively, in the following:

$$\begin{aligned} \boldsymbol{a} &= \arg\min_{\boldsymbol{h}_a} d(\boldsymbol{g}, \mathbf{W}^\top \boldsymbol{h}_a) \\ &\quad \text{s.t. } \boldsymbol{h}_a \in \mathbf{P_i} \text{ and } \boldsymbol{h}_a \in \text{Class}(\boldsymbol{f}) \end{aligned} \tag{6}$$

$$\begin{aligned} \boldsymbol{b} &= \arg\min_{\boldsymbol{h}_b} d(\boldsymbol{g}, \mathbf{W}^\top \boldsymbol{h}_b) \\ &\quad \text{s.t. } \boldsymbol{h}_b \in \mathbf{P_i} \text{ and } \boldsymbol{h}_b \notin \text{Class}(\boldsymbol{f}) \end{aligned} \tag{7}$$

where $\boldsymbol{h}_a$ and $\boldsymbol{h}_b$, respectively, refer to samples in the same-class prototype sets and the different-class prototype sets, and $d(\cdot)$ computes the Euclidean distance. Then the projections of the NNs $\boldsymbol{a}$ and $\boldsymbol{b}$ are computed utilizing the following:

$$\boldsymbol{T}_W(\boldsymbol{g}) = \mathbf{W}^\top \boldsymbol{a} \quad \text{and} \quad \boldsymbol{T}_b(\boldsymbol{g}) = \mathbf{W}^\top \boldsymbol{b}. \tag{8}$$

Step 2: For each sample $\boldsymbol{f}$, it should be more closer to the spectral–spatial prototypes of its own class than that of different classes after linear projection $\mathbf{W}$. To achieve this goal, a sigmoid function shown in (9) is employed to calculate

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the loss considering the differential property for optimization, where Q_f is defined as (10) and β denotes the parameter

$$S_\beta(Q_f) = \frac{1}{1+e^{\beta(1-Q_f)}} \tag{9}$$

$$Q_f = \frac{d(\boldsymbol{g}, \boldsymbol{T}_W(\boldsymbol{g}))}{d(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))}. \tag{10}$$

Utilizing (9) and (10), if $\boldsymbol{f}$ is closer to the spectral–spatial prototypes of the same class, the loss would be very small. On the contrary, if $\boldsymbol{f}$ is closer to the spectral–spatial prototypes of the different classes, the loss would be very large.

Step 3: Finally, the objective loss function can be written as follows:

$$J(\mathbf{W}, \mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_N) = \sum_{i=1}^{N} \sum_{f \in \mathbf{F}_i} S_\beta(Q_f). \tag{11}$$

The optimization problem of spectral–spatial prototype learning, which aims to learn the optimal spectral–spatial prototypes set $\mathcal{P} = \{\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_N\}$ and the discriminative linear projection $\mathbf{W}$, can be modeled as follows:

$$\{\mathbf{W}^*, \mathbf{P}_1^*, \ldots, \mathbf{P}_N^*\} = \underset{\mathbf{W}, \mathbf{P}_1, \ldots, \mathbf{P}_N}{\arg\min}\ J(\mathbf{W}, \mathbf{P}_1, \ldots, \mathbf{P}_N). \tag{12}$$

The learned spectral–spatial prototype set and the linear projection by solving the optimization problem in (12) would make each test local spectral–spatial set to be accurately classified to the same class with its NN spectral–spatial prototypes set, which can improve the classification accuracy significantly under the small sample size situations.

3) Optimization: To solve the optimization problem of the spectral–spatial prototype learning in (12) and obtain the optimal spectral–spatial prototypes set $\mathcal{P} = \{\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_N\}$ and the linear projection $\mathbf{W}$, a gradient descent method is employed. In the optimization, it is assumed that if the variation in the projection matrix and the spectral–spatial prototype set is sufficiently small, the same prototype neighbor is searched. The gradient decent method needs to, respectively, compute the gradients, the step size, and the path, whose details are described as follows.

Step 1: Compute the gradient of the loss function with respect to the projection. We approximately derive the gradient of J with respect to $\mathbf{W}$ based on the chain rule as follows:

$$\frac{\partial J}{\partial \mathbf{W}} = \sum_{i=1}^{N} \sum_{f \in \mathbf{F}_i} \frac{S'_\beta(Q_f)}{d^2(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))} \frac{\partial d(\boldsymbol{g}, \boldsymbol{T}_W(\boldsymbol{g}))}{\partial \mathbf{W}} d(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g})) - \sum_{i=1}^{N} \sum_{f \in \mathbf{F}_i} \frac{S'_\beta(Q_f)}{d^2(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))} d(\boldsymbol{g}, \boldsymbol{T}_w(\boldsymbol{g})) \frac{\partial d(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))}{\partial \mathbf{W}} \tag{13}$$

where $((\partial d(\boldsymbol{g}, \boldsymbol{T}_W(\boldsymbol{g})))/\partial \mathbf{W})$ and $((\partial d(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g})))/\partial \mathbf{W})$ can be easily calculated in the following, respectively:

$$\frac{\partial d(\boldsymbol{g}, \boldsymbol{T}_W(\boldsymbol{g}))}{\partial \mathbf{W}} = \frac{(\boldsymbol{f} - \boldsymbol{a})(\boldsymbol{g} - \boldsymbol{T}_W(\boldsymbol{g}))}{d(\boldsymbol{g}, \boldsymbol{T}_W(\boldsymbol{g}))} \tag{14}$$

$$\frac{\partial d(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))}{\partial \mathbf{W}} = \frac{(\boldsymbol{f} - \boldsymbol{b})(\boldsymbol{g} - \boldsymbol{T}_b(\boldsymbol{g}))}{d(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))}. \tag{15}$$

So the gradient of J with respect to $\mathbf{W}$ can be rewritten by substituting (14) and (15) into (13) as follows:

$$\frac{\partial J}{\partial \mathbf{W}} = \sum_{i=1}^{N} \sum_{f \in \mathbf{F}_i} \frac{S'_\beta(Q_f) Q_f}{\mathrm{d}^2(\boldsymbol{g}, \boldsymbol{T}_W(\boldsymbol{g}))} (\boldsymbol{f} - \boldsymbol{a})(\boldsymbol{g} - \boldsymbol{T}_W(\boldsymbol{g})) - \sum_{i=1}^{N} \sum_{f \in \mathbf{F}_i} \frac{S'_\beta(Q_f) Q_f}{\mathrm{d}^2(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))} (\boldsymbol{f} - \boldsymbol{b})(\boldsymbol{g} - \boldsymbol{T}_b(\boldsymbol{g})). \tag{16}$$

Step 2: Compute the gradient of the loss function with the spectral–spatial prototypes set. Since learning spectral–spatial prototypes set $\mathbf{P}_i, i = 1, 2, \ldots, N$ can be transmitted to learn the corresponding $\boldsymbol{v}_i^j$, $j = 1, 2, \ldots, K$, we compute the gradient of J with respect to each vector $\boldsymbol{v}_i^j$ in the following:

$$\frac{\partial J}{\partial \boldsymbol{v}_i^j} = \sum_{i=1}^{N} \sum_{\substack{f \in \mathbf{F}_i \\ \boldsymbol{p}_i^j = \boldsymbol{a}}} \frac{S'_\beta(Q_f)}{d^2(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))} \frac{\partial d(\boldsymbol{g}, \boldsymbol{T}_W(\boldsymbol{g}))}{\partial \boldsymbol{v}_i^g} d(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g})) - \sum_{i=1}^{N} \sum_{\substack{f \in \mathbf{F}_i \\ \boldsymbol{p}_i^j = b}} \frac{S'_\beta(Q_f)}{d^2(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))} d(\boldsymbol{g}, \boldsymbol{T}_w(\boldsymbol{g})) \frac{\partial d(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))}{\partial \boldsymbol{v}_i^g} \tag{17}$$

where $(\partial d(\boldsymbol{g}, \boldsymbol{T}_W(\boldsymbol{g})))/(\partial \boldsymbol{v}_i^j)$ and $(\partial d(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g})))/(\partial \boldsymbol{v}_i^j)$ can be easily calculated in the following, respectively:

$$\frac{\partial d(\boldsymbol{g}, \boldsymbol{T}_W(\boldsymbol{g}))}{\partial \boldsymbol{v}_i^j} = \frac{\mathbf{U}_i^\top \mathbf{W} \mathbf{W}^\top (\boldsymbol{a} - \boldsymbol{f})}{d(\boldsymbol{g}, \boldsymbol{T}_W(\boldsymbol{g}))} \tag{18}$$

$$\frac{\partial d(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))}{\partial \boldsymbol{v}_i^j} = \frac{\mathbf{U}_i^\top \mathbf{W} \mathbf{W}^\top (\boldsymbol{b} - \boldsymbol{f})}{d(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))}. \tag{19}$$

So the gradient of J with respect to $\boldsymbol{v}_i^j$, $j = 1, 2, \ldots, K$ can be rewritten by substituting (18) and (19) into (17) as follows:

$$\frac{\partial J}{\partial \boldsymbol{v}_i^j} = \sum_{i=1}^{N} \sum_{\substack{f \in \mathbf{F}_i \\ \boldsymbol{p}_i^j = a}} \frac{S'_\beta(Q_f) Q_f}{\mathrm{d}^2(\boldsymbol{g}, \boldsymbol{T}_W(\boldsymbol{g}))} \mathbf{U}_i^\top \mathbf{W} \mathbf{W}^\top (\boldsymbol{a} - \boldsymbol{f}) - \sum_{i=1}^{N} \sum_{\substack{f \in \mathbf{F}_i \\ \boldsymbol{p}_i^j = b}} \frac{S'_\beta(Q_f) Q_f}{\mathrm{d}^2(\boldsymbol{g}, \boldsymbol{T}_b(\boldsymbol{g}))} \mathbf{U}_i^\top \mathbf{W} \mathbf{W}^\top (\boldsymbol{b} - \boldsymbol{f}). \tag{20}$$

Step 3: Compute the step size $\boldsymbol{\tau}$. Based on the gradients computed earlier, the step size $\boldsymbol{\tau}$ is controlled by utilizing the limited-memory BFGS (L-BFGS) method.

Step 4: The path of gradient decent. In this article, the vector $\boldsymbol{v}_i^j$ is updated along straight lines based on the gradient and the step size in each iteration by employing L-BFGS. While the linear projection $\mathbf{W}$ is updated along a curved path by employing the curvilinear searching method, where the curve $\mathbf{W}(\boldsymbol{\tau})$ is calculated in the following:

$$\mathbf{W}(\boldsymbol{\tau}) = \mathbf{W} - \boldsymbol{\tau} \mathbf{A} \left(\frac{\mathbf{W} + \mathbf{W}(\boldsymbol{\tau})}{2} \right) \tag{21}$$

where $\mathbf{A}$ is computed by the following:

$$\mathbf{A} = \frac{\partial J}{\partial \mathbf{W}} \mathbf{W}^\top - \mathbf{W} \left(\frac{\partial J}{\partial \mathbf{W}} \right)^\top. \tag{22}$$

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Step 5: After the iterative procedure of gradient decent method, we can obtain the optimal spectral–spatial prototypes set and the discriminative linear projection.

The whole algorithm of the spectral–spatial prototype learning is shown in Algorithm 1.

Algorithm 1 Spectral–Spatial Prototype Learning

Require:
Training Spectral–spatial sets $\mathcal{F} = [\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_N]$ and their labels; Parameter β.
Ensure:
Compute the initial prototype sets $\mathcal{P} = [\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_N]$
Compute the initial linear projection $\mathbf{W}$
while not converged **do**
  for $i = 1, 2, \ldots, N$ **do**
    for $\boldsymbol{f} \in \mathbf{F}_i$ **do**
      Compute $\boldsymbol{g} = \mathbf{W}^\top \boldsymbol{f}$
      Compute $\boldsymbol{T}_W(\boldsymbol{g})$ and $\boldsymbol{T}_b(\boldsymbol{g})$ by (9)
      Compute the loss function $J(\mathbf{W}, \mathbf{P}_1, \ldots, \mathbf{P}_N)$
      Compute the gradients $\frac{\partial J}{\partial \mathbf{W}}$ and $\frac{\partial J}{\partial \boldsymbol{v}_i^j}$ by (17) and (21) respectively
    end for
  end for
  Compute the step size $\boldsymbol{\tau}$ by L-BFGS
  Compute $\mathbf{A}$ by (23)
  Compute the curve $\mathbf{W}(\boldsymbol{\tau})$ by curvilinear searching algorithm
  Update $\mathbf{W}$ and $\mathbf{V}_i$ by L-BFGS
end while
Compute the optimal spectral–spatial prototype sets $\mathbf{P}_i, i = 1, 2, \ldots, N$ by $\mathbf{V}_i, i = 1, 2, \ldots, N$
Output: the optimal spectral–spatial prototypes $\mathbf{P}_i, i = 1, 2, \ldots, N$ and the discriminative projection $\mathbf{W}$

C. Spectral–Spatial Prototype Learning-Based NN Classifier

After the spectral–spatial prototype training, the spectral–spatial prototype set $\mathcal{P} = [\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_N]$ for the training spectral–spatial sets $\mathcal{F} = [\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_N]$ are obtained to optimally explore both the similarity and variance of pixels within each spectral–spatial set and excavate the unseen spectral–spatial variations, which can offer more complementary information for better classification performance. Also, a discriminative linear projection $\mathbf{W}$ is achieved to make each local spectral–spatial set to be optimally classified to the same class with its NN spectral–spatial prototypes set in the projected target subspace. So, we employ the simplest NN classifier based on measuring the minimum geometric distance between the projected test spectral–spatial set and the optimal projected spectral–spatial prototypes set to determine the label. The main steps are described as follows.

Step 1: Compute the projection of the spectral–spatial prototypes $\mathbf{P}_i, i = 1, 2, \ldots, N$ of these training spectral–spatial sets and the test spectral–spatial set Y constructed for each test sample based on the linear projection $\mathbf{W}$ as follows:

$$\mathbf{P}_i^* = \mathbf{W}\mathbf{P}_i, \quad i = 1, 2, \ldots, N, \text{ and } \mathbf{Y}^* = \mathbf{W}\mathbf{Y}. \tag{23}$$

Step 2: Compute the geometric distances between the projected test spectral–spatial set and the optimal projected spectral–spatial prototype sets in the target subspace to characterize the distance between the test spectral–spatial set and the training spectral–spatial sets based on the following:

$$D\left(\mathbf{Y}^*, \mathbf{P}_i^*\right) = \|\mathbf{Y}^* - \mathbf{P}_i^*\|^2, \quad i = 1, 2, \ldots, N. \tag{24}$$

Step 3: Determine the label of the test spectral–spatial set by assigning the label of the spectral–spatial prototype set with the minimum geometric distance according to the following:

$$\text{Class}(\mathbf{Y}) = \underset{i=1,2,\ldots N}{\arg\min}\, D\left(\mathbf{Y}^*, \mathbf{P}_i^*\right). \tag{25}$$

The detailed steps of the proposed SSPLNN method for HSI classification are presented in Algorithm 2.

Algorithm 2 SSPLNN for HSI Classification

Require:
The original HSI dataset $\mathcal{X}$; Parameters r; L; M.
Ensure:
Construct the local spectral–spatial set $\mathbf{F}_i = [\boldsymbol{f}_i^1, \ldots, \boldsymbol{f}_i^M]$ for each sample;
Randomly select N samples and constitute the training spectral–spatial sets $\mathcal{F} = [\mathbf{F}_1, \mathbf{F}_2, \ldots, \mathbf{F}_N]$;
Compute the spectral–spatial prototype sets $\mathbf{P}_i, i = 1, 2, \ldots, N$ for the training spectral–spatial sets and the linear projection $\mathbf{W}$ based on the spectral–spatial prototype learning model shown in Algorithm 1;
for each test spectral–spatial set $\mathbf{Y}$ **do**
  Compute the projection of $\mathbf{P}_i, i = 1, 2, \ldots, N$ and $\mathbf{Y}$ by (24);
  Compute the geometric distances between $\mathbf{Y}^*$ and $\mathbf{P}_i^*, i = 1, 2, \ldots, N$ in the target subspace by (25);
  Determine the label of $\mathbf{Y}$ by (26);
end for
Output: The result classification map

IV. Experimental Results and Analysis

In this section, the classification performance of the proposed SSPLNN for HSI is evaluated using three benchmark HSI datasets. A brief description of the three benchmark HSI datasets is first presented. Then, the main parameters are analyzed for the proposed SSPLNN method. Next, ablation analysis is conducted to prove the effectiveness of various strategies in our proposed method. In the end, the comparison to various state-of-the-art classification approaches is performed to verify the effectiveness and the superiorities of the proposed SSPLNN.

A. Dataset Description

The three utilized HSI datasets are, respectively, Indian Pines, University of Pavia, and Salinas, whose false color images and ground truth maps are, respectively, shown in Fig. 2–4. The information of ground-truth classes of these three HSI datasets are all listed in Table I. The brief description

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TABLE I

NUMBERS OF TRAINING SAMPLES AND TESTING SAMPLES ON THE THREE HSI DATASETS

Indian Pines				Pavia University				Salinas			
Class	Name	Train	Test	Class	Name	Train	Test	Class	Name	Train	Test
1	Alfalfa	5	41	1	Asphalt	5	6626	1	Broccoli-green-weeds 1	5	2004
2	Corn-no till	5	1423	2	Meadows	5	18644	2	Broccoli-green-weeds 2	5	3721
3	Corn-min till	5	825	3	Gravel	5	2094	3	Fallow	5	1971
4	Corn	5	232	4	Trees	5	3059	4	Fallow-rough-plow	5	1386
5	Grass/Pasture	5	478	5	Metrl sheet	5	1340	5	Fallow-smooth	5	2673
6	Grass/Trees	5	725	6	Grass/Trees	5	5024	6	Stubble	5	3954
7	Grass/Pasture-mowed	5	24	7	Bitumen	5	1325	7	Celery	5	3574
8	Hay-windrowed	5	473	8	Brick	5	3677	8	Grapes-untrained	5	11266
9	Oats	5	15	9	Shadows	5	942	9	Soil-vinyary-develop	5	6198
10	Soybeans-no till	5	987					10	Corn-senesced-green-weeds	5	3273
11	Soybeans-min till	5	2450					11	Lettuce-romaine-4wk	5	1063
12	Soybeans-clean till	5	588					12	Lettuce-romaine-5wk	5	1922
13	Wheat	5	200					13	Lettuce-romaine-6wk	5	911
14	Woods	5	1260					14	Lettuce-romaine-7wk	5	1065
15	Bldg-grass-tree-drives	5	381					15	Vineyard-untrained	5	7263
16	Stone-steel towers	5	88					16	Vineyard-vertical-trellis	5	1802
	Total	80	10169		Total	45	42731		Total	80	54049

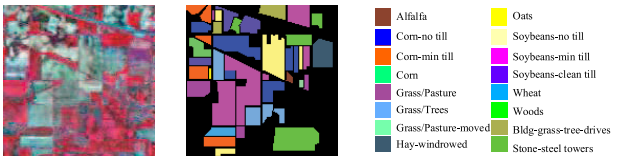


Fig. 2. Indian Pines. (a) False color image. (b) Ground-truth map. (c) Name of each class.

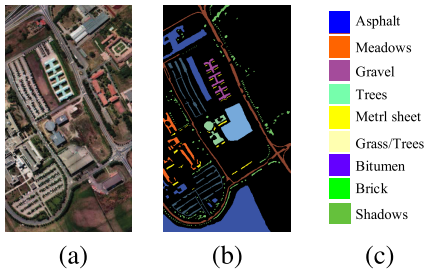


Fig. 3. University of Pavia. (a) False color image. (b) Ground-truth map. (c) Name of each class.

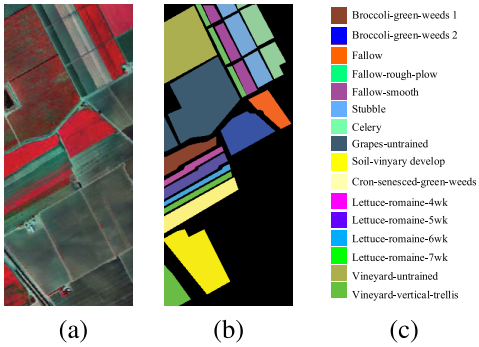


Fig. 4. Salinas. (a) False color image. (b) Ground-truth map. (c) Name of each class.

of the three benchmark HSI datasets is first presented as follows.

1) *Indian Pines:* The Indian Pines dataset was collected by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor at the northwestern Indiana in 1992. The image size is 145 × 145 × 220, and 16 ground-truth classes are included in the dataset. The spatial resolution is 20 m and the spectral range is 0.4–2.5 μm. In the classification experiments, 20 noisy spectral bands are discarded.
2) *University of Pavia:* The University of Pavia dataset from Northern Italy was acquired for the urban area using the reflective optics system imaging spectrometer (ROSIS) sensor. The image size is 610 × 340 × 115, and nine ground-truth classes are included in the dataset. The spatial resolution is 1.3 m and the spectral range is 0.43–0.86 μm. Twelve noisy spectral bands are removed for the classification experiments.
3) *Salinas:* The Salinas dataset from the Salinas Valley in California was acquired by the ROSIS sensor. The image size is 512 × 217 × 224, and 16 ground-truth classes are included in the dataset. The spatial resolution is 3.7 m. For the classification task, a total of 20 water absorption bands are discarded.

B. Experimental Configuration

In the following experiments, only five labeled samples per class are randomly chosen to construct the training sets for all the three HSI datasets, and the remaining samples are utilized as test. The number of training samples and the number of test samples of the three HSI datasets are listed in Table I. Four evaluation indexes, i.e., overall accuracy (OA), average accuracy (AA), Kappa coefficient, and accuracy of each class, are employed to quantitatively appraise the classification performance. Particularly, the classification results in this article are all averaged over ten independent experiments.

In addition, the compared handcrafted feature extraction (FE)-based classification methods and the proposed method are all implemented in MATLAB 2014a on a computer with a 3.2-GHz Intel Core i5 processor and 8.0-GB memory with a Windows 10 operating system, and the compared deep-learning-based classification methods are implemented on the same computer equipped with a single 1060 Ti GPU.

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C. Analysis of the Reduced Spectral Dimension r

The reduced spectral dimension r determines the spectral size of the local spectral–spatial neighbor set, which is one of the most important parameters in the proposed method. To analyze the impact of r for classification performance, we change r from 5 to 50 and present the classification results of all the three HSI datasets in Fig. 5, including OA (%), AA (%), and Kappa (%).

It can be observed from Fig. 5 that the OA, AA, and Kappa coefficient increase rapidly with the reduced spectral dimension r ranging from 5 to 20 for all the three HSI datasets. While with r continue increasing from 20 to 50, the classification accuracy OA, AA, and Kappa coefficient show slight decrease for the three datasets. The main reason is that more and more main spectral information in the original HSI dataset would be retained when the reduced spectral dimension r increase from 5 to 20. However, as r continues to grow, some noise information would be introduced to reduce the classification accuracy. Moreover, the higher the spectral dimension r is, the higher the computational complexity of the proposed method. Considering the classification performance, the reduced spectral dimension r is fixed as 20 in our proposed method for all the three HSI datasets.

D. Analysis of the Number of Superpixels

Since the number of superpixel L has a great influence on the extraction of spatial structure information, we analyze the impact of L in our proposed method for classification performance. In the following experiments, the number of superpixels L increases from 30 to 120 in steps of 10 for Indian Pines and Salinas, from 100 to 300 in steps of 20 for Pavia University. The classification results in terms of OA (%), AA (%), and Kappa coefficient (%) of all the three HSI datasets are shown in Fig. 6.

From Fig. 6, we observe that the OA, AA, and Kappa coefficient all increase with the number of superpixels L changing from 30 to 80 for Indian Pines and Salinas, and from 100 to 160 for Pavia University. When L continues to grow, the classification accuracy in terms of OA, AA, and Kappa all decrease. If the number of superpixels L is too large, the size of each individual superpixel is too small, which cannot fully explore the spatial structure information for classification. On the contrary, if the number of superpixels L is too small, the size of each individual superpixel is too large, which is very likely to introduce some dissimilar pixels especially in edge areas. Considering the classification performance, the number of superpixels L is optimally fixed as 80 in our proposed method for Indian Pines and Salinas, and as 160 for Pavia University.

E. Analysis of the Number of Local Spectral–Spatial Neighbors M

The number of local spectral–spatial neighbors M is one of the most important parameters determining the spatial size of the local spectral–spatial neighbor set in our proposed method. We conduct a series of experiments when the number of local spectral–spatial neighbors M increases from 100 to 300 for all the three HSI datasets to verify the effects of M in our proposed method. The classification results in terms of OA (%), AA (%), and Kappa coefficient (%) of all the three HSI datasets with M changing are shown in Fig. 7.

TABLE II
OPTIMAL VALUES FOR PARAMETERS OF THE PROPOSED METHOD SSPLNN

Parameters	Indian Pines	Pavia University	Salinas
r	20	20	20
L	80	160	80
M	200	160	200

It can be observed from Fig. 7 that the classification accuracy in terms of OA, AA, and Kappa all grow when the number of local spectral–spatial neighbors M increases from 100 to 200 for Indian Pines and Salinas, and from 100 to 160 for Pavia University. With M further increasing, the classification accuracy decreases slightly for all the three datasets. The main reason is that the local spatial information within each individual superpixel cannot be adequately exploited when M is small, while the dissimilar pixel would be introduced when M is large. Considering the classification performance, the number of local spectral–spatial neighbors M is optimally fixed as 200 in our proposed method for Indian Pines and Salinas, and as 160 for Pavia University.

To sum up, the most important parameters in the proposed method are all fixed as the values indicated in Table II, including the reduced spectral dimension r, the number of superpixel L, and the number of local spectral–spatial neighbors M. In addition, the parameter β, i.e., the slope of the sigmoid function, is set as 1 for all the three datasets.

F. Ablation Analysis

Ablation experiment and analysis are conducted on three widely used HSI datasets in this section to prove the effectiveness of various strategies in our proposed method. The compared methods include the KNN classifier, prototype learning without local spectral–spatial neighbor set (PLNN), and spectral–spatial prototype learning without discriminative projection (SSPLNW). Four indexes are utilized to evaluate the classification performance, i.e., OA, AA, the Kappa coefficient, and the accuracy of each class. The mean classification results with standard variance (in percentage) based on KNN, PLNN, SSPLNW, SSPLNN for Indian Pines, University of Pavia, and Salinas are, respectively, listed in Tables III–V.

From Tables III–V, we observe that the proposed SSPLNN achieves the highest classification accuracy for all the three datasets. Also, the experimental results show that our methods of image processing in various stages are effective in improving the classification accuracy to some extent. The classification accuracy of PLNN is much higher than that of KNN for all the three datasets, which indicates that our designed spectral–spatial prototype learning model can excavate more unseen discriminative spectral–spatial features to improve the classification accuracy remarkably under small sample size situations. Moreover, SSPLNW obtains better

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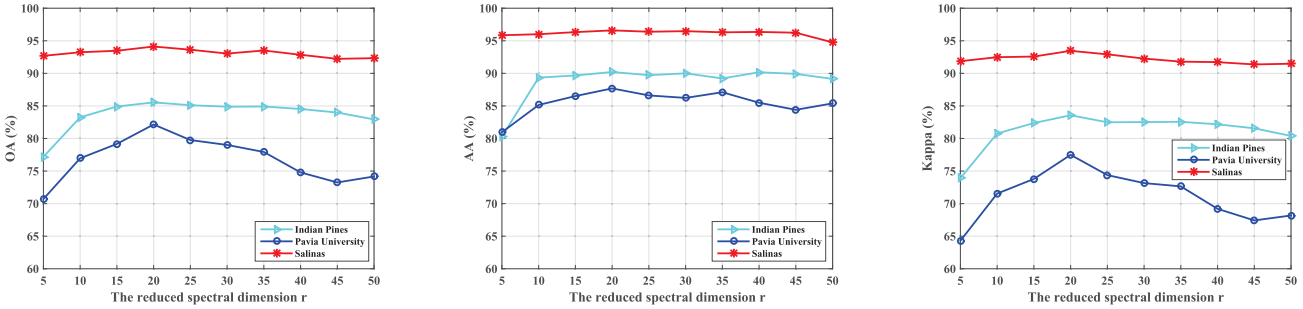


Fig. 5. Impacts of the reduced spectral dimension r in our proposed method for HSI classification. (a) OA (%) results. (b) AA (%) results. (c) Kappa (%) results.

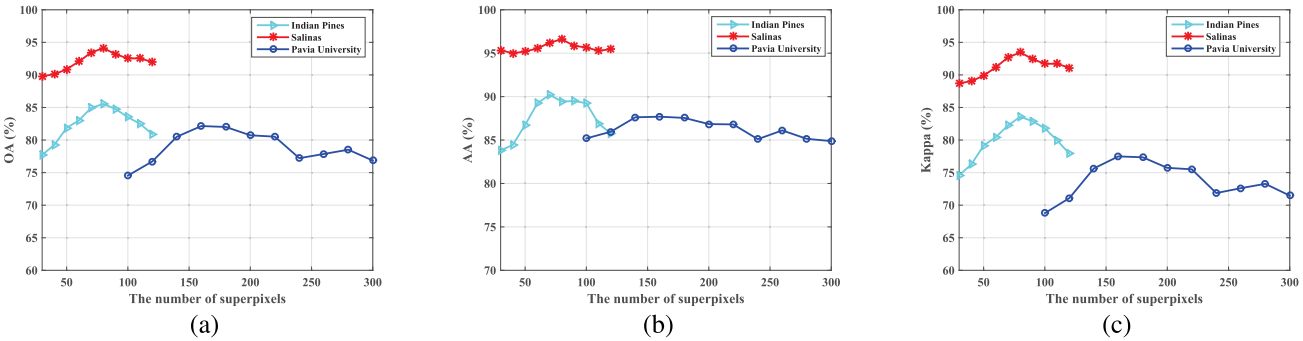


Fig. 6. Impacts of the number of superpixels in our proposed method for HSI classification. (a) OA (%) results. (b) AA (%) results. (c) Kappa (%) results.

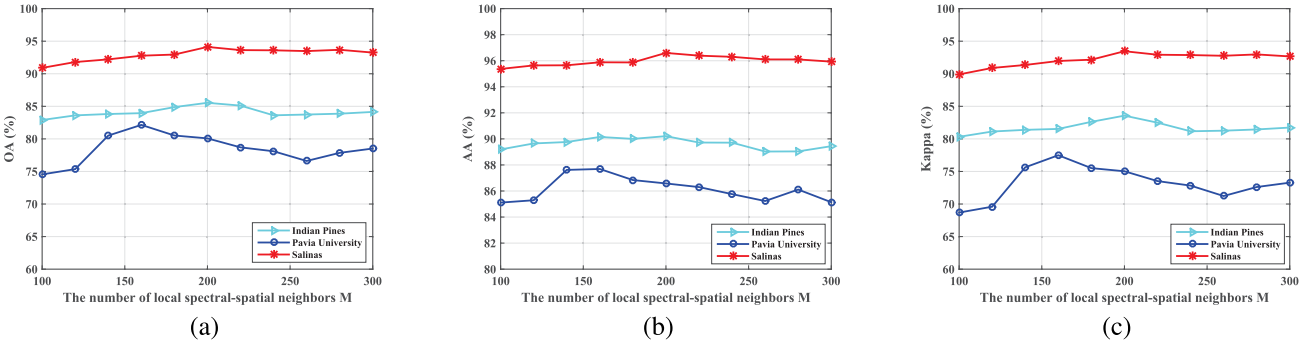


Fig. 7. Impacts of the number of local spectral–spatial neighbors M in our proposed method for HSI classification. (a) OA (%) results. (b) AA (%) results. (c) Kappa (%) results.

classification performance than PLNN. The main reason is that local spectral–spatial set is constructed for each sample in SSPLNW to explore the local spectral–spatial information accurately. In addition, the fact that the classification accuracy of our proposed SSPLNN is superior to that of SSPLNW demonstrates that the learned linear discriminative projection by the spectral–spatial prototype learning model can further enhance the classification performance. Tables III–V show that the classification accuracies of some classes obtained by SSPLNW (such as the 4th, 6th, 10th, 12th, 15th classes of Indian Pines, the 9th class of Pavia University, and the 1st, 2nd, 6th, 15th, 16th classes of Salinas) are higher than that obtained by SSPLNN. The major reason is that the designed spectral–spatial prototype learning model in SSPLNN learns a linear discriminative projection while HSI datasets are complex with a large number of nonlinearly separable samples.

To further verify the influence of each method of image processing in various stages for the computational complexity of the proposed SSPLNN method, we conduct experiments to report the computational time cost by each proposed strategy. Table VI shows the computational time of all the three HSI datasets, respectively, cost by the construction of local spectral–spatial set (i.e., the preprocessing stage), the spectral–spatial prototype learning model, and the NN classifier based on the geometric distance between the projected test spectral–spatial set and the optimal projected spectral–spatial prototype sets. Moreover, the computational time based on the set-to-set NN classifier for the three HSI datasets is listed in Table VI for comparison.

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TABLE III

CLASSIFICATION RESULTS OF KNN, PLNN, SSPLNW, AND SSPLNN FOR INDIAN PINES

Class	KNN	PLNN	SSPLNW	SSPLNN
1	82.44 ± 7.95	83.17 ± 18.23	**100.00** ± 0.00	**100.00** ± 0.00
2	46.09 ± 7.05	57.67 ± 12.61	72.91 ± 10.07	**78.38** ± 11.81
3	56.32 ± 8.41	55.99 ± 16.72	56.91 ± 16.98	**68.05** ± 20.46
4	59.87 ± 10.91	92.33 ± 8.39	**97.20** ± 6.29	95.09 ± 8.54
5	74.62 ± 5.74	73.56 ± 18.20	69.21 ± 7.08	**75.36** ± 8.14
6	70.44 ± 11.28	66.44 ± 6.34	**99.86** ± 0.00	98.73 ± 2.38
7	93.48 ± 11.64	66.96 ± 44.16	90.00 ± 3.61	**100.00** ± 0.00
8	65.71 ± 14.32	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00
9	99.33 ± 2.11	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00
10	52.18 ± 12.84	79.29 ± 6.07	**85.32** ± 7.98	85.12 ± 8.22
11	45.82 ± 8.49	79.91 ± 7.06	74.49 ± 7.75	**83.84** ± 9.50
12	48.91 ± 11.91	64.76 ± 14.66	**88.21** ± 13.74	80.03 ± 13.73
13	97.40 ± 2.07	99.85 ± 0.24	**100.00** ± 0.00	**100.00** ± 0.00
14	77.74 ± 10.45	88.20 ± 11.49	92.71 ± 5.67	**97.31** ± 3.91
15	50.00 ± 8.24	82.20 ± 13.02	**93.83** ± 9.71	82.13 ± 19.08
16	86.82 ± 3.87	90.23 ± 4.46	97.73 ± 0.00	**97.84** ± 0.36
OA	57.66 ± 2.96	75.54 ± 2.52	81.79 ± 3.26	**85.54** ± 1.96
AA	69.20 ± 1.32	80.16 ± 3.96	88.65 ± 2.21	**90.12** ± 1.19
Kappa (%)	52.57 ± 3.23	72.49 ± 2.80	79.47 ± 3.61	**85.54** ± 2.19

TABLE IV

CLASSIFICATION RESULTS OF KNN, PLNN, SSPLNW, AND SSPLNN FOR PAVIA UNIVERSITY

Class	KNN	PLNN	SSPLNW	SSPLNN
1	47.28 ± 14.61	70.75 ± 10.53	53.82 ± 5.88	**74.97** ± 6.64
2	53.17 ± 13.66	67.02 ± 9.96	56.02 ± 11.31	**77.18** ± 4.97
3	77.87 ± 12.61	**86.68** ± 10.04	74.73 ± 10.73	84.28 ± 9.38
4	81.93 ± 7.15	81.28 ± 6.08	66.32 ± 7.06	**85.68** ± 7.83
5	**99.85** ± 0.00	99.52 ± 0.62	99.53 ± 0.10	99.63 ± 0.38
6	77.13 ± 8.97	92.18 ± 7.93	75.97 ± 7.76	**93.72** ± 5.49
7	95.59 ± 11.05	98.92 ± 2.93	90.77 ± 8.017	**99.49** ± 0.98
8	75.75 ± 12.38	79.64 ± 10.05	60.43 ± 12.50	**86.24** ± 5.17
9	64.58 ± 13.09	79.20 ± 7.19	**94.21** ± 3.25	88.03 ± 3.29
OA	63.32 ± 6.72	75.90 ± 5.17	63.34 ± 4.63	**82.15** ± 2.94
AA	74.79 ± 3.90	83.91 ± 2.80	74.65 ± 1.55	**87.69** ± 2.07
Kappa (%)	55.73 ± 7.10	70.26 ± 5.91	55.65 ± 4.71	**77.49** ± 3.51

TABLE V

CLASSIFICATION RESULTS OF KNN, PLNN, SSPLNW, AND SSPLNN FOR SALINAS

Class	KNN	PLNN	SSPLNW	SSPLNN
1	97.84 ± 1.65	99.92 ± 0.14	**100.00** ± 0.00	99.03 ± 1.89
2	96.94 ± 0.91	99.24 ± 1.67	**100.00** ± 0.00	99.08 ± 1.02
3	84.91 ± 9.59	99.98 ± 0.04	**100.00** ± 0.00	**100.00** ± 0.00
4	95.23 ± 2.53	94.45 ± 2.81	97.75 ± 0.55	**98.82** ± 0.96
5	95.65 ± 3.50	**99.03** ± 0.17	98.71 ± 0.08	98.68 ± 0.32
6	99.59 ± 0.19	99.73 ± 0.05	**99.86** ± 0.05	**99.86** ± 0.09
7	94.91 ± 15.57	**99.93** ± 0.03	99.92 ± 0.02	99.92 ± 0
8	55.62 ± 15.73	71.50 ± 14.51	75.60 ± 15.65	**85.56** ± 4.96
9	98.84 ± 0.53	**100.00** ± 0.00	99.99 ± 0.02	99.97 ± 0.01
10	75.41 ± 13.08	92.44 ± 8.59	89.20 ± 10.12	**93.21** ± 2.15
11	92.06 ± 2.69	95.33 ± 2.51	94.52 ± 3.19	**98.25** ± 2.88
12	93.41 ± 10.76	97.88 ± 4.25	96.66 ± 3.64	**98.17** ± 4.84
13	**98.64** ± 0.52	96.86 ± 4.71	94.85 ± 7.77	98.59 ± 0.73
14	**93.45** ± 2.33	**89.07** ± 4.13	89.90 ± 4.66	91.05 ± 4.98
15	62.71 ± 10.96	85.77 ± 8.90	**87.88** ± 5.93	85.60 ± 4.87
16	87.95 ± 7.19	98.47 ± 2.77	**99.76** ± 0.39	99.73 ± 0.44
OA	81.63 ± 3.03	90.93 ± 2.09	91.97 ± 2.72	**94.12** ± 0.64
AA	88.95 ± 1.45	94.98 ± 0.68	95.29 ± 0.99	**96.60** ± 0.50
Kappa (%)	79.62 ± 3.30	89.94 ± 2.29	91.09 ± 2.98	**93.46** ± 0.70

Table VI shows that the NN classifier based on the geometric distance between the projected test spectral–spatial set and the optimal projected spectral–spatial prototype sets in our proposed SSPLNN takes the most computational time. The main reason is that each test sample in HSI dataset needs to be compared with the prototype set of every training sample. Nevertheless, the computational time cost by the NN classifier in our proposed SSPLNN is much less than that cost by the set-to-set NN classifier, which indicates that the NN classifier based on spectral–spatial prototype learning can improve the computational speed.

TABLE VI

COMPUTATIONAL TIME COST BY EACH STRATEGY OF THE PROPOSED SSPLNN

Strategies	Indian Pines	Pavia University	Salinas
Set-to-set NN classifier	669.69	2508.79	5705.53
Preprocessing stage	2.59	34.69	35.43
Prototype learning	54.75	32.17	59.38
NN classifier of SSPLNN	145.21	782.03	883.35

G. Comparison to the State-of-the-Art Algorithms

To verify the effectiveness of the proposed method, SSPLNN is compared to several advanced classification methods, including the local covariance matrix representation (LCMR) [67], the HybridSN [36], the similarity-based deep metric model (S-DMM)-based classification method [68], the dual graph convolutional network with drop edge classifier (DGCN-dc) [69], the semi-supervised classification via spatial-regulated self-training (Semi-S) [70], the semi-supervised cross-scale graph prototypical network (X-GPN) [56], and the dual-channel capsule generation adversarial network (DcCaps-GAN) [71]. Here, LCMR is the latest and advanced hand-craft FE-based classification methods, whereas HybridSN, S-DMM, GGCN-dc, Semi-S, X-GPN, and DcCapsGAN are the latest and advanced deep-learning-based classification methods. In addition, most of the compared methods, such as LCMR, HybridSN, S-DMM, GGCN-dc, and Semi-S, are all specifically developed for HSI classification under small sample size situation. Moreover, Semi-S and X-GPN are semi-supervised classification methods, and X-GPN is an advanced classification method based on prototype learning and deep learning. To achieve the best performance, the parameters in the compared LCMR, HybridSN, S-DMM, DGCN-dc, Semi-S, X-GPN, and DcCapsGAN are all set to their optimal values.

In this section, the numbers of the training and testing samples of all the classification methods in the following experiments are set as the same as shown in Table I. The quantitative mean classification results with standard variance (in percentage) of all the eight methods for Indian Pines, Pavia University, and Salinas are, respectively, reported in Tables VII–IX, and the classification maps obtained by these methods with some corresponding zoomed-in parts are displayed in Fig. 8–10, respectively.

From Tables VII–IX, we observe that the proposed method SSPLNN achieves the best classification results. Moreover, SSPLNN obtains nearly 7% higher OA value than the other seven methods for Indian Pines, nearly 5% higher OA value for Salinas, and nearly 3% higher OA value for Pavia University, which demonstrate that SSPLNN is obviously superior to the compared state-of-the-art methods. Although some of deep-learning-based classification methods, such as HybridSN, S-DMM, DGCN-dc, and Semi-S, are spectrally developed for HSI classification under small sample size situation and automatically extract discriminative spectral–spatial features to improve the classification performance, SSPLNN

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TABLE VII

CLASSIFICATION RESULTS OF LCMR, HYBRIDSN, S-DMM, DGCN-DC, SEMI-S, X-GPN, DCCAPSGAN, AND THE PROPOSED SSPLNN FOR INDIAN PINES

Method	LCMR [67]	HybridSN [36]	S-DMM [68]	DGCN-DC [69]	Semi-S [70]	X-GPN [56]	DcCapsGAN [71]	SSPLNN
OA	74.21 ± 2.06	63.89 ± 3.00	59.83 ± 3.65	78.25 ± 1.93	76.27 ± 3.04	67.16 ± 4.17	66.71 ± 0.08	**85.54** ± 1.96
AA	85.10 ± 1.40	77.20 ± 1.80	73.01 ± 1.52	87.56 ± 1.10	86.50 ± 1.61	81.06 ± 1.41	73.55 ± 0.11	**90.12** ± 1.19
Kappa (%)	71.03 ± 2.16	59.57 ± 3.35	55.06 ± 3.53	75.51 ± 2.11	73.29 ± 3.36	62.93 ± 4.46	62.18 ± 0.05	**83.57** ± 2.19
1	99.02 ± 1.26	99.30 ± 1.12	92.44 ± 4.11	99.23 ± 1.73	99.75 ± 0.79	98.39 ± 2.08	**100.00** ± 0.00	**100.00** ± 0.00
2	63.18 ± 11.20	49.51 ± 16.28	33.11 ± 13.66	69.04 ± 9.83	67.12 ± 13.74	52.48 ± 15.46	49.98 ± 0.28	**78.38** ± 11.81
3	60.06 ± 11.87	39.52 ± 14.88	38.11 ± 16.52	63.86 ± 9.18	58.96 ± 9.60	57.07 ± 17.89	44.66 ± 0.14	**68.05** ± 20.46
4	87.33 ± 11.11	84.32 ± 24.98	61.72 ± 25.73	93.43 ± 8.06	94.59 ± 4.91	88.92 ± 12.60	47.01 ± 0.52	**95.09** ± 8.54
5	84.39 ± 11.54	68.40 ± 22.93	72.13 ± 13.00	**89.43** ± 9.16	83.82 ± 10.35	81.70 ± 7.48	64.98 ± 0.23	75.36 ± 8.14
6	88.84 ± 8.36	75.76 ± 16.79	87.89 ± 6.36	87.57 ± 7.22	83.92 ± 6.81	95.89 ± 3.11	84.02 ± 0.11	**98.73** ± 2.38
7	**100.00** ± 0.00	99.60 ± 1.26	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	98.52 ± 4.67	**100.00** ± 0.00	**100.00** ± 0.00
8	**100.00** ± 0.00	95.28 ± 5.58	84.65 ± 7.68	98.62 ± 3.11	95.53 ± 5.37	90.11 ± 11.58	79.95 ± 0.19	**100.00** ± 0.00
9	96.00 ± 7.17	97.65 ± 7.44	**100.00** ± 0.00	**100.00** ± 0.00	**100.00** ± 0.00	99.15 ± 2.29	**100.00** ± 0.00	**100.00** ± 0.00
10	65.19 ± 10.64	58.92 ± 14.28	58.77 ± 10.92	74.28 ± 11.77	76.24 ± 7.18	63.64 ± 12.74	62.45 ± 0.35	**85.12** ± 8.22
11	60.01 ± 10.20	51.59 ± 11.46	52.67 ± 19.68	66.86 ± 6.63	62.69 ± 6.68	50.26 ± 14.65	64.83 ± 0.21	**83.84** ± 9.50
12	**80.15** ± 13.01	48.80 ± 14.27	45.07 ± 13.21	78.98 ± 8.37	74.77 ± 7.34	66.01 ± 15.92	49.54 ± 0.41	80.03 ± 13.73
13	**100.00** ± 0.00	98.37 ± 2.95	98.95 ± 1.38	99.49 ± 0.24	98.99 ± 1.09	96.88 ± 7.17	98.99 ± 0.06	**100.00** ± 0.00
14	91.04 ± 6.78	91.87 ± 5.94	84.25 ± 11.26	93.24 ± 4.68	94.96 ± 5.50	74.15 ± 16.82	94.75 ± 0.26	**97.31** ± 3.91
15	91.50 ± 12.09	78.83 ± 15.62	59.71 ± 15.75	91.19 ± 11.05	**95.47** ± 7.00	85.07 ± 11.28	36.89 ± 0.82	82.13 ± 19.08
16	94.89 ± 2.09	97.56 ± 3.84	**98.64** ± 1.50	95.70 ± 4.90	97.13 ± 3.76	98.69 ± 3.42	98.76 ± 0.36	97.84 ± 0.36
Computational time (Sec.)	12.73	73.29	541.72	1429.38	1329.92	1291.77	2431.68	198.91

TABLE VIII

CLASSIFICATION RESULTS OF LCMR, HYBRIDSN, S-DMM, DGCN-DC, SEMI-S, X-GPN, DCCAPSGAN, AND THE PROPOSED SSPLNN FOR PAVIA UNIVERSITY

Method	LCMR [67]	HybridSN [36]	S-DMM [68]	DGCN-DC [69]	Semi-S [70]	X-GPN [56]	DcCapsGAN [71]	SSPLNN
OA	77.25 ± 5.40	65.35 ± 5.77	73.67 ± 4.40	79.05 ± 5.45	72.00 ± 4.48	72.50 ± 5.26	70.09 ± 1.12	**82.15** ± 2.94
AA	82.10 ± 3.99	70.91 ± 3.50	81.25 ± 4.07	85.41 ± 3.60	76.87 ± 3.96	79.18 ± 2.71	78.88 ± 0.71	**87.69** ± 2.07
Kappa (%)	71.11 ± 6.37	57.10 ± 5.94	67.31 ± 5.30	73.89 ± 6.25	64.86 ± 4.54	65.61 ± 5.83	63.15 ± 1.14	**77.49** ± 3.51
1	74.65 ± 14.04	59.06 ± 9.11	70.28 ± 8.22	**80.13** ± 5.89	74.28 ± 9.37	61.38 ± 13.95	36.58 ± 5.33	74.97 ± 6.64
2	75.12 ± 9.00	63.54 ± 11.56	67.22 ± 9.13	72.46 ± 11.10	67.15 ± 15.91	70.10 ± 12.70	75.02 ± 0.68	**77.18** ± 4.97
3	74.31 ± 13.33	59.99 ± 10.23	77.14 ± 7.34	83.70 ± 5.44	69.25 ± 16.69	75.59 ± 10.41	78.09 ± 1.49	**84.28** ± 9.38
4	96.32 ± 3.34	88.41 ± 4.00	93.45 ± 6.69	84.65 ± 2.44	89.96 ± 5.23	81.88 ± 5.26	**98.30** ± 0.09	85.68 ± 7.83
5	96.06 ± 8.64	84.39 ± 7.14	93.58 ± 5.99	95.53 ± 7.41	84.14 ± 24.38	99.87 ± 0.19	**100.00** ± 0.00	99.63 ± 0.38
6	75.61 ± 11.40	65.49 ± 10.64	75.79 ± 11.42	87.13 ± 7.15	82.82 ± 22.20	74.13 ± 15.88	53.98 ± 0.41	**93.72** ± 5.49
7	89.76 ± 8.58	80.84 ± 9.90	89.91 ± 8.98	91.89 ± 5.15	77.73 ± 17.26	96.47 ± 3.93	97.96 ± 0.57	**99.49** ± 0.98
8	65.83 ± 17.84	52.06 ± 8.76	73.74 ± 10.71	69.72 ± 13.12	67.06 ± 15.78	71.55 ± 16.22	70.01 ± 3.13	**86.24** ± 5.17
9	91.23 ± 5.41	84.37 ± 10.41	90.11 ± 5.59	92.58 ± 2.80	89.42 ± 2.47	81.72 ± 11.82	**100.00** ± 0.00	88.03 ± 3.29
Computational time (Sec.)	121.17	141.39	2169.69	673.15	1675.36	775.06	1088.61	846.84

TABLE IX

CLASSIFICATION RESULTS OF LCMR, HYBRIDSN, S-DMM, DGCN-DC, SEMI-S, X-GPN, DCCAPSGAN, AND THE PROPOSED SSPLNN FOR SALINAS

Method	LCMR [67]	HybridSN [36]	S-DMM [68]	DGCN-DC [69]	Semi-S [70]	X-GPN [56]	DcCapsGAN [71]	SSPLNN
OA	88.90 ± 1.97	87.08 ± 2.82	87.79 ± 2.74	87.34 ± 2.71	89.94 ± 1.83	86.27 ± 2.24	82.15 ± 0.20	**94.12** ± 0.64
AA	93.07 ± 1.62	90.98 ± 2.34	93.28 ± 1.44	91.86 ± 1.40	93.77 ± 1.49	90.71 ± 0.91	91.87 ± 0.17	**96.60** ± 0.50
Kappa (%)	87.65 ± 2.19	85.65 ± 3.11	86.45 ± 3.02	85.95 ± 3.00	88.82 ± 2.02	84.71 ± 2.43	81.06 ± 0.21	**93.46** ± 0.70
1	99.62 ± 1.10	**99.91** ± 0.14	98.96 ± 2.09	98.61 ± 4.23	99.85 ± 0.23	98.79 ± 3.04	96.98 ± 2.16	99.03 ± 1.89
2	82.07 ± 11.68	83.29 ± 4.63	98.23 ± 2.93	81.37 ± 12.31	82.50 ± 11.84	98.30 ± 3.21	**99.94** ± 0.11	99.08 ± 1.02
3	98.86 ± 1.85	99.28 ± 0.93	94.04 ± 7.62	96.52 ± 3.25	98.81 ± 1.76	95.11 ± 13.83	**100.00** ± 0.00	**100.00** ± 0.00
4	99.98 ± 0.05	99.92 ± 0.20	99.17 ± 1.25	**99.99** ± 0.05	98.03 ± 4.18	99.20 ± 0.85	99.79 ± 0.06	98.82 ± 0.96
5	95.44 ± 2.35	93.36 ± 5.62	97.15 ± 3.61	96.19 ± 1.01	96.35 ± 0.69	91.72 ± 8.19	96.42 ± 0.22	**98.68** ± 0.32
6	98.92 ± 0.93	98.19 ± 0.67	99.38 ± 1.16	98.57 ± 1.19	98.93 ± 0.93	98.26 ± 3.05	**100.00** ± 0.00	**99.86** ± 0.09
7	96.20 ± 2.21	80.97 ± 22.88	99.22 ± 1.96	91.86 ± 14.41	94.99 ± 8.77	96.20 ± 9.81	99.75 ± 0.18	**99.92** ± 0.00
8	82.20 ± 6.84	83.48 ± 6.98	67.35 ± 10.61	75.55 ± 9.40	80.25 ± 10.82	67.38 ± 14.88	62.77 ± 0.48	**85.56** ± 4.96
9	90.59 ± 7.65	78.60 ± 14.99	97.98 ± 2.45	87.72 ± 11.39	92.60 ± 6.79	99.51 ± 1.27	99.88 ± 0.05	**99.97** ± 0.01
10	91.50 ± 10.78	78.82 ± 14.16	83.44 ± 10.60	86.62 ± 12.79	91.49 ± 8.30	79.89 ± 14.92	79.65 ± 0.23	**93.21** ± 2.15
11	99.96 ± 0.05	99.95 ± 0.05	96.12 ± 4.01	**99.95** ± 0.05	99.85 ± 0.41	99.93 ± 0.08	98.10 ± 0.17	98.25 ± 2.88
12	91.92 ± 10.34	92.16 ± 14.55	97.74 ± 6.34	93.34 ± 6.88	95.79 ± 9.11	90.62 ± 6.38	**99.86** ± 0.06	98.17 ± 4.84
13	96.23 ± 5.18	97.72 ± 1.40	**99.66** ± 0.35	96.62 ± 2.49	97.76 ± 1.27	93.45 ± 12.47	99.62 ± 0.04	98.59 ± 0.73
14	95.11 ± 3.84	**96.56** ± 3.29	94.37 ± 6.10	96.00 ± 3.07	95.13 ± 4.15	93.19 ± 6.85	99.51 ± 0.09	91.05 ± 4.98
15	75.06 ± 8.40	84.42 ± 10.43	77.73 ± 7.37	83.04 ± 8.55	83.51 ± 10.48	70.95 ± 15.74	40.09 ± 1.72	**85.60** ± 4.87
16	95.38 ± 4.99	89.07 ± 7.08	91.88 ± 6.11	87.81 ± 10.34	94.49 ± 6.37	94.65 ± 6.17	97.64 ± 0.10	**99.73** ± 0.44
Computational time (Sec.)	189.47	182.73	2492.38	1741.75	1583.79	1378.59	2598.43	969.95

shows much higher classification accuracy than them. Also, SSPLNN achieves much better classification performance than X-GPN embedding prototype learning. The main reason is that SSPLNN learns the spectral–spatial prototypes, which can not only utilize both the similarity and variance of the pixels within each spectral–spatial set but also excavate more unseen complementary spectral–spatial variations. These spectral–spatial information extracted by the spectral–spatial prototypes is much effective to improve the classification accuracy significantly under the small sample size situation. Also, SSPLNN simultaneously learns a linear discriminative projection to make each test local spectral–spatial set in the projected target subspace to be optimally classified to the same class with its NN spectral–spatial prototype set, which can further enhance the classification performance. Therefore, the proposed SSPLNN method offers a significant improvement for HSI classification performance with more robustness under small sample size situation. The superior classification performance of the proposed SSPLNN method can also be illustrated from the classification maps displayed in Figs. 8–10.

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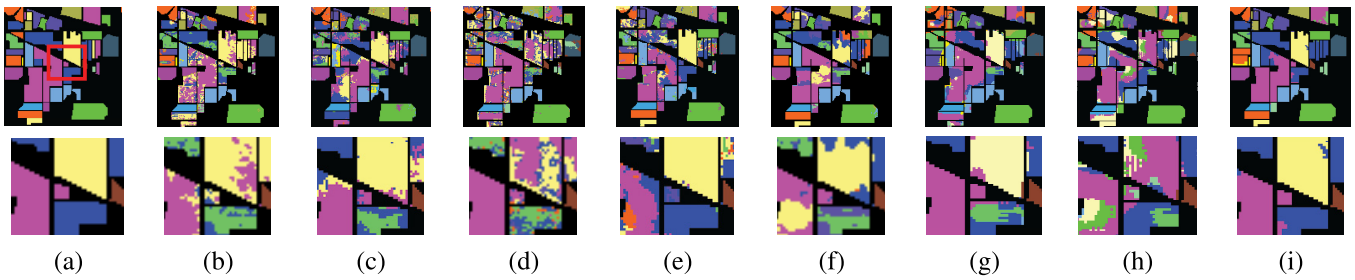


Fig. 8. Classification maps with their corresponding zoomed-in parts in the red box obtained by the state-of-the-art classification methods for Indian Pines. (a) Ground-truth. (b) LCMR. (c) HybridSN. (d) S-DMM. (e) DGCN-DC. (f) Semi-S. (g) X-GPN. (h) DcCapsGAN. (i) proposed SSPLNN.

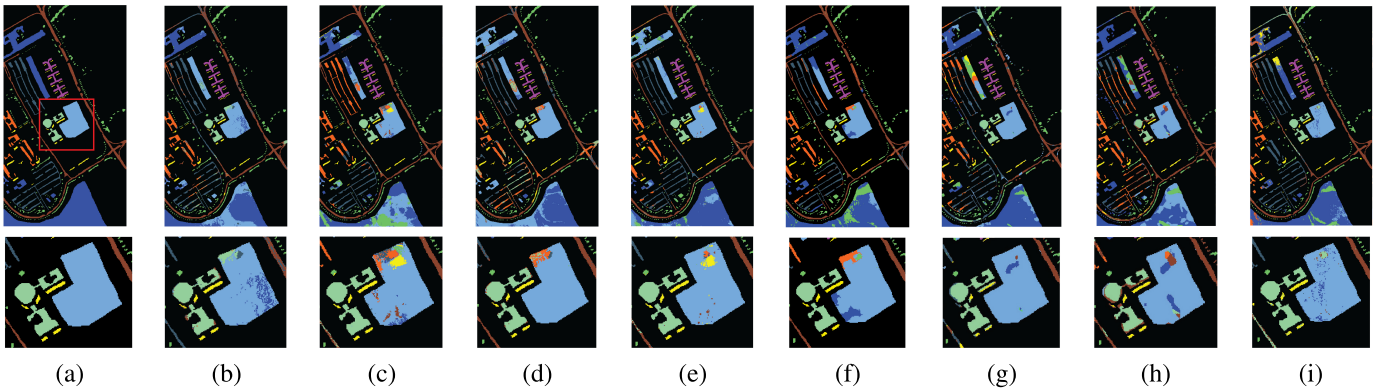


Fig. 9. Classification maps with their corresponding zoomed-in parts in the red box obtained by the state-of-the-art classification methods for Pavia University. (a) Ground-truth. (b) LCMR. (c) HybridSN. (d) S-DMM. (e) DGCN-DC. (f) Semi-S. (g) X-GPN. (h) DcCapsGAN. (i) proposed SSPLNN.

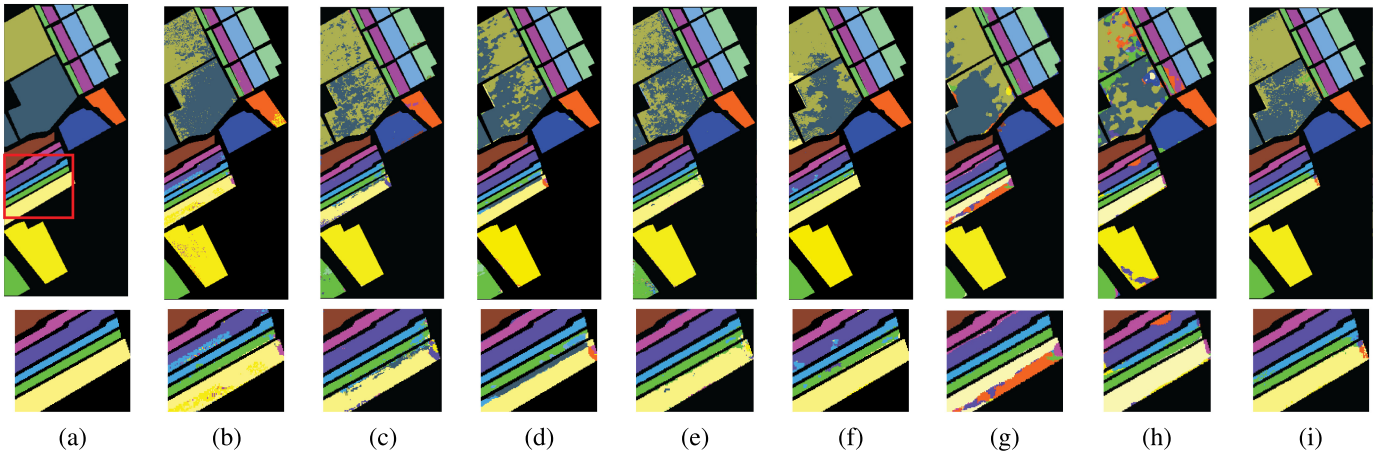


Fig. 10. Classification maps with their corresponding zoomed-in parts in the red box obtained by the state-of-the-art classification methods for Salinas. (a) Ground-truth. (b) LCMR. (c) HybridSN. (d) S-DMM. (e) DGCN-dc. (f) Semi-S. (g) X-GPN. (h) DcCapsGAN. (i) proposed SSPLNN.

It can also be observed from Tables VII–IX that the computational speed of SSPLNN is faster than that of S-DMM, DGCN-dc, Semi-S, X-GPN, and DcCapsGAN but slower than that of LCMR and HybridSN for all the three HSI datasets. The main reason is that most deep-learning-based methods contain a large number of parameters to be tuned, which leads to much higher computational complexity. The spectral–spatial prototype learning and the NN classifier based on the geometric distance between the projected test spectral–spatial set and the optimal projected spectral–spatial prototype sets in the proposed SSPLNN both need high computational costs. To sum up, considering both the classification accuracy and the computational speed, the proposed SSPLNN method outperforms various latest and advanced classification methods by a large margin.

H. Analysis of the Number of Training Samples

Finally, we conduct experiments to analyze the effect of the number of training samples for classification to further highlight the advantages of the proposed SSPLNN. The number of training samples per class is, respectively, set as 3, 7, and 9 for all the three HSI datasets and the classification results in terms of OA are, respectively, reported in Fig. 11(a)–(c).

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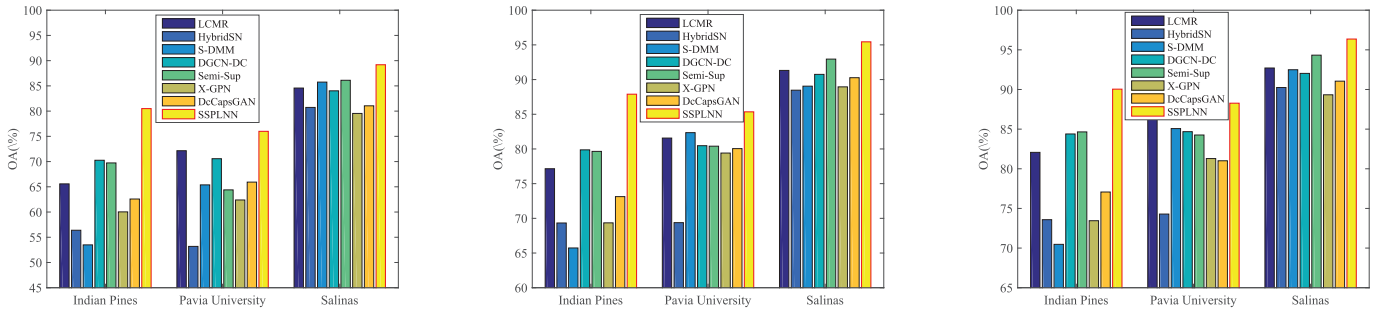


Fig. 11. Classification results by LCMR, HybridSN, S-DMM, DGCN-DC, Semi-S, X-GPN, DcCapsGAN, and the proposed SSPLNN with the number of training samples per class are set as 3, 7, and 9 for all the three datasets. (a) Classification results with 3 training samples per class. (b) Classification results with 7 training samples per class. (c) Classification results with 9 training samples per class.

From Fig. 11, we can find that the proposed SSPLNN achieves the best classification performance under all the different numbers of training sample situations. Moreover, the smaller the number of training samples is, the greater that advantages of the proposed SSPLNN method over the compared classification methods. These results demonstrate that the proposed SSPLNN outperforms the state-of-the-art classification methods with a large margin especially under the small sample size situations.

V. Conclusion

This article proposes a novel spectral–spatial prototype learning-based NN classifier for HSI. The local spectral–spatial neighbor set is first constructed for each pixel to accurately explore the local spectral–spatial information. Then, a spectral–spatial prototype learning model is designed to simultaneously learn a set of spectral–spatial prototypes and a discriminative linear projection. The learned spectral–spatial prototypes are utilized to excavate more unseen spectral–spatial information. The linear projection is utilized to make each test sample to be optimally classified to the same class in the projected target subspace. The simplest NN classifier by measuring the minimum geometric distance between the projected test spectral–spatial set and the optimal projected spectral–spatial prototype sets is employed to complete the classification task. The experimental results on three widely analyzed HSI datasets demonstrate that the proposed SSPLNN outperforms various latest and advanced classification methods by a large margin under the small sample size situation. However, there are still some disadvantages in our proposed method. One is that the NN classifier based on prototype learning has relatively high computational complexity. The other is that the proposed method contains many parameters that need to be analyzed and optimized. In our further work, we would like to design more advanced classifiers combining with the spectral–spatial prototype learning model to improve the computational speed and try to construct a learnable parameter module based on deep learning method to avoid the manual calculation of parameter optimization.

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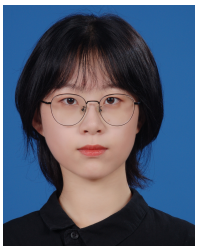
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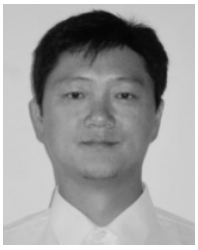
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